

7 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

The 0.22 moles will turn into vapor, whose pressure is called **vapor pressure or partial vapor pressure**:

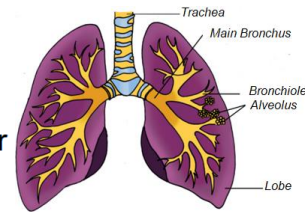
$$P_{\text{vapor}} V = nRT \Rightarrow P_{\text{vapor}} = \frac{nRT}{V} = \frac{0.22 \times 8.31 \times (273 + 100)}{10 \times 10^{-3}} \simeq 68252 \text{ Pa} = 0.68 \text{ bar}$$

Remarks

- ↪ The evaporation process persists as long as the **vapor pressure** (P_{vapor}) remains lower than the **saturation pressure** (P_{sat}).
- ↪ **Equilibrium** between the liquid and gas phases is reached when $P_{\text{vapor}} = P_{\text{sat}}$, from then on, no more vapor molecules can be formed because the enclosure is then **saturated**.

Medical Application: Respiratory Water Loss and Humidity

The human body eliminates 2.4l of water (urine, sweat, etc.) per day, of which 300 to 400g are eliminated by the **lungs** via the alveoli (**evaporation**). However, this mass of water vapor depends on the **humidity of the inspired air**. Indeed:



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- ↪ If the inspired air is **saturated** with water vapor, the elimination of water from the body by the alveoli will be **zero** (evaporation is impossible since P_{sat} is already reached).
- ↪ If the inspired air is **dry**, the elimination will be **maximal**.

	Composition of inspired air	Composition of expired air
N_2	79%	79%
O_2	21%	16%
CO_2	0.03%	4.5%
Vapor of H_2O	Variable quantity	Very abundant

D- THE SOLUTIONS

1- DEFINITIONS

- ↪ A solution is a **homogeneous mixture**, i.e., it consists of a **single phase**.

Example: dissolution of sugar (solute) in water (solvent).

- ↪ **Colloidal solution**: a **heterogeneous mixture**, i.e., it consists of more than one phase.

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The solute particles (colloids) are relatively large, with sizes ranging from 10 to 1000 Å.

Example: dispersion of soap in water.

↪ **Electrolytic solution:** an ionic solution that is electrically neutral and conducts electric current.

Example: NaCl solution.

↪ **Ideal solution:** a solution in which *solvent – solvent*, *solute – solute*, and *solute – solvent* interactions are approximately equal, meaning that no single interaction predominates.

Solutions are considered **ideal** when they are **very dilute** ($C \leq 10^{-3} \text{ mol/l}$).

Example: a mixture of alcohols.

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2- CHEMICAL ACTIVITY

In a solution, the interactions (*solute* ↔ *solvent*) and (*solute* ↔ *solute*) affect the **availability** of the solute for any chemical reaction.

Therefore, the notion of **active concentration** (a_i) - also called **chemical activity** - is introduced for species "i" (atom, molecule, or ion) with concentration C_i :

$$a_i = \gamma_i \frac{C_i}{C_0} \quad a_i : \text{dimensionless.}$$

$C_0 = 1 \text{ mol/l}$: reference concentration.

γ_i : **activity coefficient**. $0 < \gamma_i \leq 1$

$\gamma_i = 1$ for **ideal solutions**, **pure substances**, and **solvents**.

3 - IONIC STRENGTH « I » OF A SOLUTION

The solution consists of k ions, each with a valence z_k and a concentration C_k . The **ionic strength** is given by:

$$I = \frac{1}{2} \sum_k C_k z_k^2 \quad [I] = \text{mol/l}$$

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Recall

$$z_{Al^{3+}} = +3; z_{O^{2-}} = -2; z_{H^+} = +1$$

↗ The **ionic strength** plays a role in the **chemical activity** of ions. For $I < 0.05 \text{ mol/l}$, this effect is expressed by the **Debye-Hückel** equation:

$$\log \gamma_i = -0,5 z_i^2 \sqrt{\frac{I}{I_0}}$$

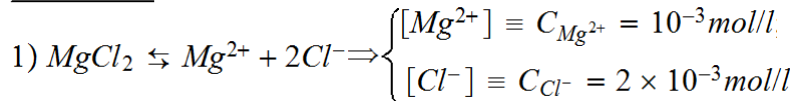
$I_0 = 1 \text{ mol/l}$: reference ionic strength.

Exercise 3:

Calculate the activity coefficient of the chloride ion in $MgCl_2$ at a concentration of 10^{-3} mol/l and in $LaCl_3$ at the same concentration. Conclude.

La : Lanthanum.

Solutions:



We calculate the ionic strength of the solution:

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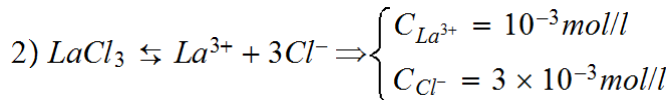
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$$\begin{aligned} I &= \frac{1}{2} \sum_k C_k z_k^2 = \frac{1}{2} [C_{Mg^{2+}} \times (+2)^2 + C_{Cl^-} \times (-1)^2] \\ &= \frac{1}{2} [10^{-3} \times (+2)^2 + 2 \times 10^{-3} \times (-1)^2] = 3 \times 10^{-3} \text{ mol/l} \end{aligned}$$

$I < 5 \times 10^{-2} \Rightarrow$ we apply the **Debye-Hückel** formula:

$$\log \gamma_i = -0,5 z_i^2 \sqrt{\frac{I}{I_0}} = -0,5 (-1)^2 \sqrt{3 \times 10^{-3}} \Rightarrow \gamma_{Cl^-} = 0.9389$$



$$I = \frac{1}{2} \sum_k C_k z_k^2 = \frac{1}{2} [C_{La^{3+}} \times (+3)^2 + C_{Cl^-} \times (-1)^2] = 6 \times 10^{-3} \text{ mol/l}$$

$$\begin{aligned} I < 5 \times 10^{-2} \Rightarrow \log \gamma_i &= -0,5 z_i^2 \sqrt{\frac{I}{I_0}} = -0,5 (-1)^2 \sqrt{6 \times 10^{-3}} \\ &\Rightarrow \gamma_{Cl^-} = 0.9147 \end{aligned}$$

Conclusion:

The activity coefficient γ_i varies with the ionic strength I , and if $I_1 < I_2 \Rightarrow \gamma_1 > \gamma_2$.

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4 - CHEMICAL POTENTIAL

The energy of a system varies with the variation in the number of moles of any species i in that system. This gives species i a potential called the **chemical potential** μ_i , which is defined by:

$$\mu_i(T) = \mu_i^0(T) + RT \ln(a_i) \quad [\mu_i] = J/mol$$

$\mu_i^0(T)$: potential of i in its **standard state**.

The **standard state** of a substance is its **most stable** physical state at P_{atm} , for any given temperature T . For example, at 298K, the standard state of water is **liquid**.

Remark

When two phases coexist at **equilibrium**, their **temperatures**, **pressures**, and **chemical potentials** are equal.

Exercise 4:

- 1) Calculate the variation of the chemical potential of glucose when, at 27°C, its concentration in water increases from $C_1 = 0.1 \text{ mmol/l}$ to $C_2 = 0.2 \text{ mmol/l}$.

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- 2) Why is the chemical potential μ of water lower in an aqueous solution than in pure water?

Solutions:

$$1) C_1 < 1 \text{ mmol/l} \Rightarrow \text{ideal solution} \Rightarrow \mu_{g1}(T) = \mu_g^0(T) + RT \ln\left(\frac{C_1}{C_0}\right)$$

$$C_2 < 1 \text{ mmol/l} \Rightarrow \text{ideal solution} \Rightarrow \mu_{g2}(T) = \mu_g^0(T) + RT \ln\left(\frac{C_2}{C_0}\right)$$

$$\Rightarrow \mu_{g2} - \mu_{g1} = RT \ln \frac{C_2}{C_1} = 8.31 \times 300 \times \ln 2 = 1730 J/mol$$

- 2) Concentration of pure water (C_1) :

There are 55.5 moles in 1l of water: $C_1 = 55.5 \text{ mol/l}$

↗ Concentration of water in aqueous solution (C_2) :

$$\text{If } V_{\text{solution}} = 1l \Rightarrow n_{\text{water}} < 55.5 \text{ moles} \Rightarrow C_2 = \frac{n_{\text{water}}}{V_{\text{solution}}} < 55.5 \text{ mol/l}$$

We have, $\mu_2 - \mu_1 = RT \ln\left(\frac{a_2}{a_1}\right)$.

$$a_1 = \gamma_{1_{\text{water}}} \frac{C_1}{C_0} \equiv \frac{C_1}{C_0} \text{ because } \gamma_{1_{\text{water}}} = 1 \text{ (it is a pure body).}$$

$$a_2 = \gamma_{2_{\text{water}}} \frac{C_2}{C_0} \equiv \frac{C_2}{C_0} \text{ because } \gamma_{2_{\text{water}}} = 1 \text{ (it is a solvent).}$$

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$$\Rightarrow \frac{a_2}{a_1} = \frac{C_2}{C_1} \Rightarrow \mu_2 - \mu_1 = RT \ln\left(\frac{C_2}{C_1}\right)$$

$$\text{However, } C_2 < C_1 \Rightarrow \mu_2 - \mu_1 < 0 \Rightarrow \mu_2 < \mu_1$$

Therefore, the chemical potential of water in solution (μ_2) is lower than that of pure water (μ_1).

5 - OSMOLARITY OF A SOLUTION

For a solution containing a single solute, the osmolarity is defined as:

$$\omega = [1 + \alpha(\beta - 1)]C$$

$$\Rightarrow \omega = \frac{n_{\text{osmoles of solute}}}{V_{\text{solution}}} : \text{osmolarity of the solution. } [\omega] = \text{osmol/l}$$

$$\Rightarrow C = \frac{n_{\text{moles of solute}}}{V_{\text{solution}}} : \text{molarity of the solution. } [C] = \text{mol/l}$$

$\Rightarrow \beta$: **number of ions** produced by the dissociation of the solute, if it is an **electrolyte**.

$\beta = 1$ if it is a **non-electrolyte** (e.g., urea and glucose) $\Rightarrow \omega = C$